

Lab 1 - Use Case Flow

Problem Statement 07: Hostel Maintenance & Issue Ticketing System

Name: J Sanjay Varma SRN: PES1UG24CS194 Section: D

Use Case ID	UC-01
Use Case Name	Log Maintenance Ticket
Primary Actor	Hostel Resident
Other Actors	Maintenance Warden (receives the ticket), Notification Service (sends the acknowledgement)
Relationship	<<include>> Validate Ticket Details
Trigger	Resident opens the hostel app and taps “Raise New Ticket”
Requirements Covered	FR-001, and partly NFR-002

Preconditions

1. Resident has logged in and has a room allotted to them.
2. The app is able to reach the server.
3. Issue types and warden queues are already set up for each block.

Postconditions

1. A new ticket is saved with status Open and a unique ticket ID.
2. The ticket appears in the warden queue for that block and the SLA timer starts.
3. The resident gets a confirmation with the ticket ID.
4. If the use case fails, nothing is saved and no ticket ID is used up.

Main Success Scenario

1. Resident taps “Raise New Ticket”.
2. System shows the ticket form. Room number is already filled in from the resident’s profile.
3. Resident picks the issue type (plumbing, electrical or carpentry) and the severity.
4. Resident types a short description and attaches up to five photos (5 MB each).
5. Resident submits the form.
6. System runs <<include>> Validate Ticket Details to check that the room number, issue type and severity are filled in and the photos are valid.
7. Validation passes.
8. System creates the ticket, gives it an ID, sets status to Open and notes the time.
9. System sends the ticket to the warden queue for that block.
10. System shows the ticket ID on screen and sends an SMS or email confirmation.
11. Use case ends successfully.

Alternate Flow 6a: Validation Fails

Happens at step 6 if a required field is missing or a photo is rejected.

6a1. System does not create the ticket and highlights the fields that are wrong, for example “Room number is required”.

6a2. Resident fixes them and submits again. Flow goes back to step 6.

6a3. If it fails three times in a row, the system keeps the entry as a draft and tells the resident to report at the hostel office. Use case ends in failure.

Alternate Flow 4a: Photo Upload Fails

Happens at step 4 if a photo is too large or the connection drops.

4a1. System shows an upload error but keeps whatever else the resident has already typed.

4a2. Resident can retry, use a smaller photo, or carry on without a photo.

4a3. If the severity is Critical and no photo was added, the ticket is marked “photo pending” for the warden. Flow goes back to step 5.